GUAN HUANG

APPLIED ML ENGINEER

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Applied ML Engineer with **4+ years** taking ML systems from design to deployment, focused on rigorous evaluation, retrieval and inference efficiency. Led teams of 4–5 to ship production LLM agents and **Railtracks**, an open-source agent framework. MSc in Applied Computing (AI), University of Toronto.

4+	15+	200+	3
YEARS OF APPLIED ML	PRODUCTION AGENTS SHIPPED	GITHUB STARS	PATENTS PENDING

EXPERIENCE

May 2023 – Sep 2025

Applied ML Engineer / Railtown AI Technology Inc.

- Led a team of five to ship Conductr, an early end-to-end **DevOps agent** across Azure, Jira and GitHub (work-item triage, stack-trace monitoring, Q&A over code and Git history). Preferred in **90% of blind pairwise comparisons** by engineers and students over a leading commercial coding assistant (two-sided binomial, $p<0.01$).
- Designed and built **Railtracks**, the company's **open-source agent framework**.
- Built AEval, an automated **AI expert system evaluation** tool (accuracy, hallucination, security, fairness), now embedded in **hundreds of deployed products**.
- Worked with Mila researchers on stack-trace NLP models and ran product user studies.

Oct 2024 – Apr 2025

Undergraduate Researcher / UBC SALT Lab – ECE Department

- **Control-flow guided test generation:** used execution paths to guide language-agnostic test generation, raising branch coverage from 88% to 93% with Llama 3.
- Proposed and led **SoC visual schematic generation with VLMs**: working with HDL hardware files through a vision-language model with an entity database and feedback loop.

May 2022 – May 2023

Data Scientist / Railtown AI Technology Inc.

- Fine-tuned a **BERT** model for root cause analysis, raising F1 from 0.71 to 0.85 on 1,200 real historical cases and outperforming the OpenAI API baseline of the time.
- Designed a **multi-agent review system** before agent frameworks existed: models score outputs against a rubric and a reviewer agent reconciles disagreements. 91% user preference in A/B testing (two-sided binomial, $p<0.001$).

SELECTED PROJECTS

Mar 2024 – Present

Creator & Maintainer / Railtracks OPEN SOURCE

- Plain-Python agent framework where each part (loop, tools, limits, run record) is an object you can read and replace. **200+ GitHub stars**, about **1,500 downloads a month**.
- Any function becomes a tool with one decorator, agents call other agents as tools, and every run is recorded and replayed locally in a visualizer.
- Profiled latency and memory overhead against LiteLLM, LangChain and the raw OpenAI SDK.

Jan 2025 – Present

ML Team Lead / FeatherRing COMMUNITY PROJECT

- Lead the Intelligence Core team of this invite-only project: a **multi-agent system that operates the computer** and understands all media types.
- Fine-tuned **Qwen3-32B** for two-stage **context compression**: select task-relevant entities, then summarize each passage and entity.
- Engineered a **steerable LLM** that takes new instructions while it generates.

Sep 2026 – Present

Author & Maintainer / EvalRing OPEN SOURCE

- **Evaluation tool** that ranks models, prompts or pipelines on your task, reproducibly.

Oct 2026 – Present

Designer & Developer / NewsRing OPEN SOURCE

- **Daily AI news agent** on Railtracks that picks only from sources it collected.

Apr 2025 – Sep 2025

Advisor/Engineer / ClickCut PARTNERING WITH NUCHAIN

- Led a group of six building an **AI video editor** that plans and edits hour-long videos.

EDUCATION

Sep 2025 – Nov 2026

MSc in Applied Computing – Artificial Intelligence / University of Toronto

Vice President, CS Graduate Student Society: represent 600+ graduate students in governance and outreach to 20+ firms

Sep 2019 – Apr 2025

BASc in Computer Engineering / University of British Columbia

Top 1% (grade 93, honours); Trek Excellence, PMC-Sierra and James Yan scholarships

PATENTS & PUBLICATIONS

PATENTS	Three pending: ticket generation, RCA, release notes and DevOps knowledge copilots.
PUBLICATIONS	*Railtracks Agentic Framework* report (co-author); *MapRing*, *LORAG*, *VSFL* (first author), 2025.

SKILLS

LANGUAGES	Python (**PyTorch, JAX**), JS/TS, C++ (CUDA), Java, Dart
MODELING	LLM fine-tuning (Qwen3, BERT), VLMs, multi-agent systems, context compression
RETRIEVAL	BM25, dense and hybrid retrieval, reciprocal rank fusion, chunking strategies, RAG
EVALUATION	Held-out splits, blind pairwise and A/B tests, ablations, bootstrap CIs, McNemar tests
INFERENCE	TTFT, throughput and latency benchmarking, prefix caching, batching, cost accounting
INFRASTRUCTURE	vLLM, Docker/K8s, AzureML, CI/CD, FastAPI, async systems, GPU programming

FULL CV	Awards, competitions, research, more projects and events: coronring.github.io/resume